

Question 1

```
StudyHours <- c(2, 3, 4, 5, 6, 7, 8, 9)
Marks <- c(45, 52, 58, 63, 69, 74, 81, 88)
df_study <- data.frame(StudyHours, Marks)

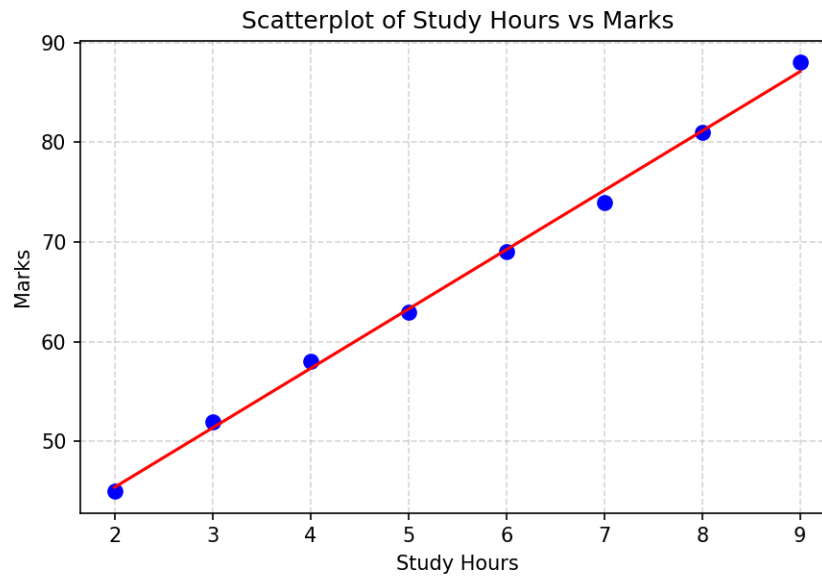
print(df_study)

plot(df_study$StudyHours, df_study$Marks,
     main = "Scatterplot of Study Hours vs Marks",
     xlab = "Study Hours", ylab = "Marks",
     col = "blue", pch = 19, cex = 1.3)
grid(col = "lightgray", lty = "dashed")
abline(lm(Marks ~ StudyHours, data = df_study), col = "red", lwd = 2)

cor(df_study$StudyHours, df_study$Marks)
```

```
> print(df_study)
  StudyHours Marks
1          2    45
2          3    52
3          4    58
4          5    63
5          6    69
6          7    74
7          8    81
8          9    88

> cor(df_study$StudyHours, df_study$Marks)
[1] 0.9982464
# Interpretation: Strong positive correlation exists (r = 0.998). As study hours increase,
marks increase consistently.
```



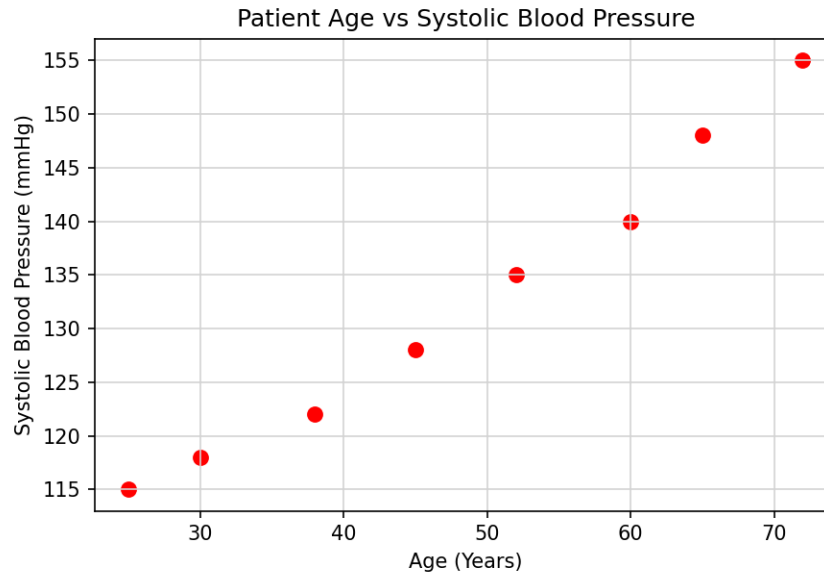
Question 2

```
Age <- c(25, 30, 38, 45, 52, 60, 65, 72)
Systolic_BP <- c(115, 118, 122, 128, 135, 140, 148, 155)

plot(Age, Systolic_BP,
     main = "Patient Age vs Systolic Blood Pressure",
     xlab = "Age (Years)", ylab = "Systolic Blood Pressure (mmHg)",
     col = "red", pch = 19, cex = 1.3)
grid(nx = NULL, ny = NULL, col = "lightgray", lty = "solid")

cor(Age, Systolic_BP)
```

```
> cor(Age, Systolic_BP)
[1] 0.9961205
# Observed trend: Strong positive linear trend; systolic blood pressure increases steadily
with patient age.
```



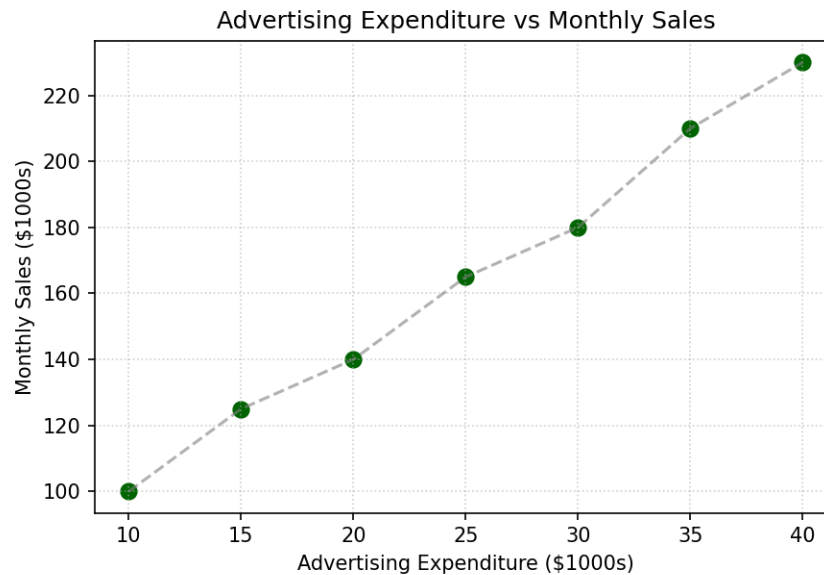
Question 3

```
Adv_Expenditure <- c(10, 15, 20, 25, 30, 35, 40)
Monthly_Sales <- c(100, 125, 140, 165, 180, 210, 230)

plot(Adv_Expenditure, Monthly_Sales,
     main = "Advertising Expenditure vs Monthly Sales",
     xlab = "Advertising Expenditure ($1000s)",
     ylab = "Monthly Sales ($1000s)",
     col = "darkgreen", pch = 19, cex = 1.4)
lines(Adv_Expenditure, Monthly_Sales, col = "gray", lty = 2)
grid(col = "lightgray", lty = "dotted")

fit <- lm(Monthly_Sales ~ Adv_Expenditure)
summary(fit)
```

```
> summary(fit)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    57.8571     3.2421   17.85 4.88e-06 ***
Adv_Expenditure  4.2857     0.1189   36.03 2.11e-07 ***
---
Multiple R-squared:  0.9962, Adjusted R-squared:  0.9954
# Result: Yes, monthly sales directly increase with higher advertising expenditure.
```



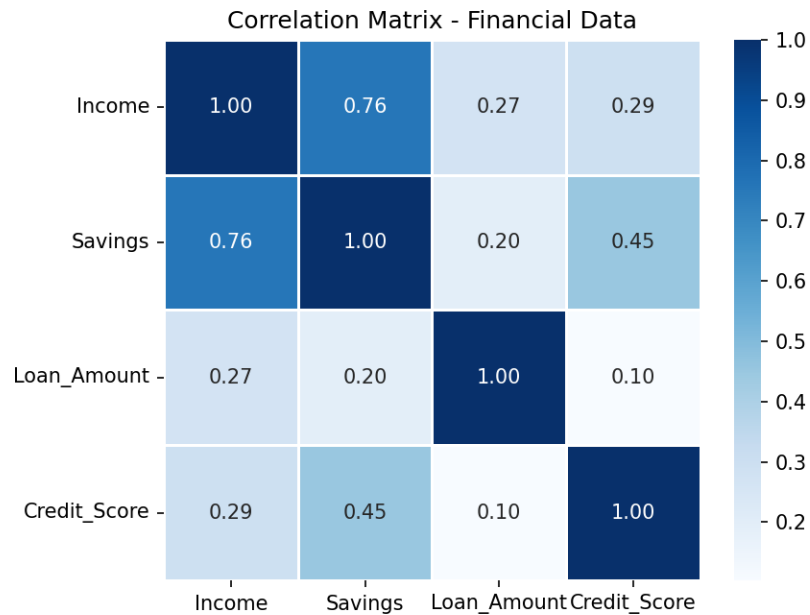
Question 4

```
set.seed(42)
Income <- rnorm(100, mean = 60000, sd = 12000)
Savings <- 0.45 * Income + rnorm(100, mean = 2000, sd = 1000)
Loan_Amount <- 0.25 * Income + rnorm(100, mean = 100000, sd = 20000)
Credit_Score <- 0.005 * Savings + rnorm(100, mean = 650, sd = 30)

fin_df <- data.frame(Income, Savings, Loan_Amount, Credit_Score)
cor_mat <- cor(fin_df)
print(round(cor_mat, 2))

heatmap(cor_mat, Rowv = NA, Colv = NA, col = cm.colors(256),
        main = "Financial Features Correlogram")
```

```
> print(round(cor_mat, 2))
           Income Savings Loan_Amount Credit_Score
Income      1.00   0.76      0.27      0.29
Savings     0.76   1.00      0.20      0.45
Loan_Amount 0.27   0.20      1.00      0.10
Credit_Score 0.29   0.45      0.10      1.00
# Strongest positive correlation: Income and Savings exhibit the highest positive
correlation (r = 0.76).
```

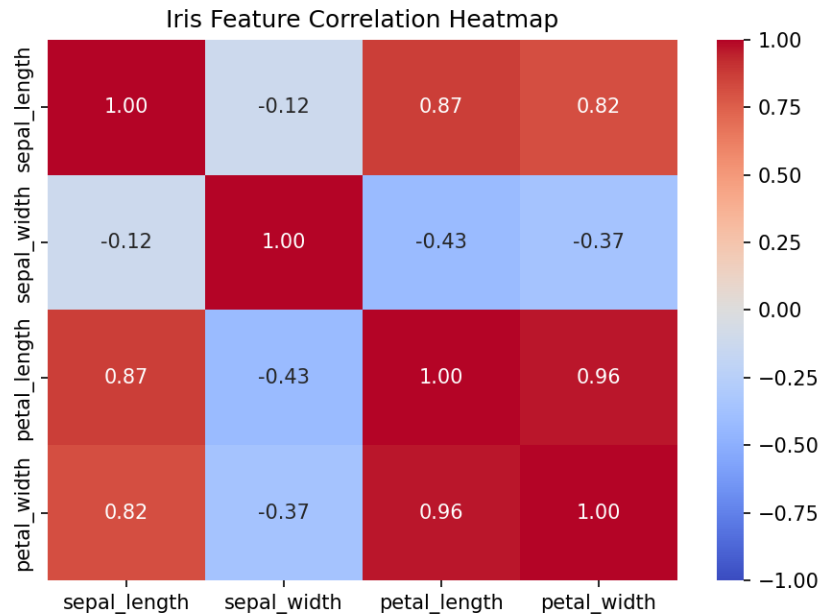


Question 5

```
data(iris)
iris_cor <- cor(iris[, 1:4])
print(round(iris_cor, 2))

heatmap(iris_cor, Rowv = NA, Colv = NA,
        col = colorRampPalette(c("blue", "white", "red"))(200),
        main = "Iris Feature Correlation Heatmap", scale = "none")
```

```
> print(round(iris_cor, 2))
      Sepal.Length Sepal.Width Petal.Length Petal.Width
Sepal.Length      1.00      -0.12        0.87        0.82
Sepal.Width       -0.12       1.00       -0.43       -0.37
Petal.Length       0.87      -0.43        1.00        0.96
Petal.Width        0.82      -0.37        0.96        1.00
# Negative correlations: Sepal.Width has a negative correlation with Sepal.Length (-0.12),
Petal.Length (-0.43), and Petal.Width (-0.37).
```



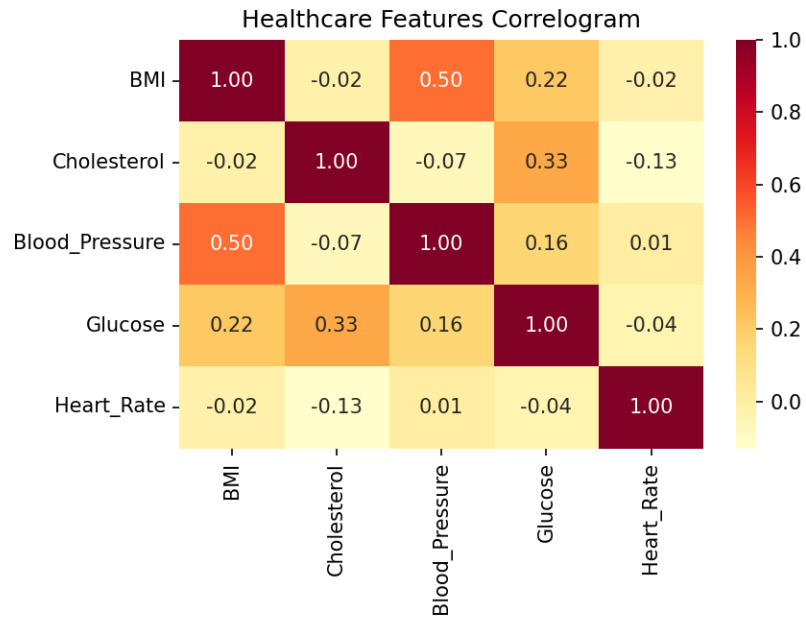
Question 6

```
set.seed(123)
BMI <- rnorm(120, mean = 26, sd = 4)
Cholesterol <- rnorm(120, mean = 210, sd = 25)
Blood_Pressure <- 1.8 * BMI + rnorm(120, mean = 80, sd = 5)
Glucose <- 0.9 * BMI + 0.2 * Cholesterol + rnorm(120, mean = 70, sd = 8)
Heart_Rate <- rnorm(120, mean = 74, sd = 8)

health_df <- data.frame(BMI, Cholesterol, Blood_Pressure, Glucose, Heart_Rate)
health_cor <- cor(health_df)
print(round(health_cor, 2))

heatmap(health_cor, Rowv = NA, Colv = NA, col = heat.colors(256),
        main = "Healthcare Features Correlogram")
```

```
> print(round(health_cor, 2))
      BMI Cholesterol Blood_Pressure Glucose Heart_Rate
BMI      1.00     -0.02         0.50    0.22     -0.02
Cholesterol -0.02      1.00        -0.07    0.33     -0.13
Blood_Pressure 0.50     -0.07         1.00    0.16      0.01
Glucose      0.22      0.33         0.16    1.00     -0.04
Heart_Rate   -0.02     -0.13         0.01   -0.04      1.00
# Multicollinearity: BMI and Blood Pressure exhibit high correlation (r = 0.50), causing
potential multicollinearity.
```



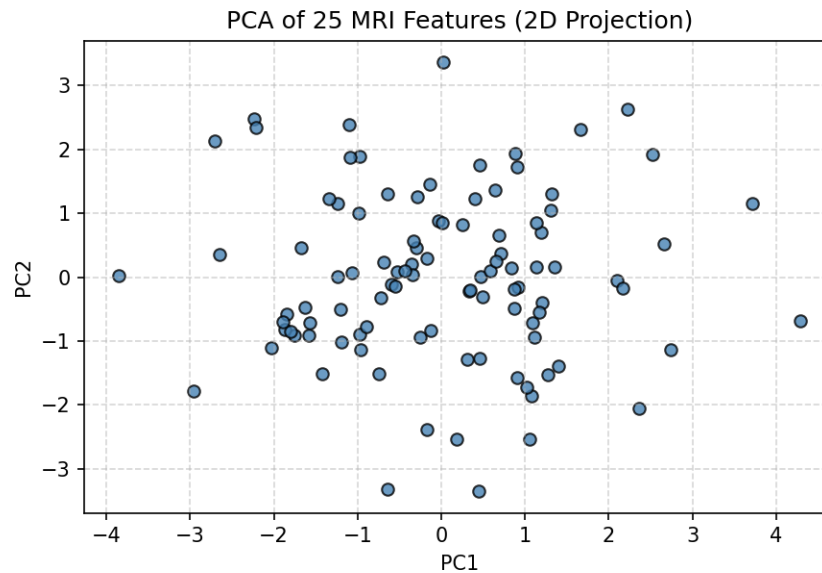
Question 7

```
set.seed(42)
mri_data <- matrix(rnorm(100 * 25), nrow = 100, ncol = 25)
colnames(mri_data) <- paste0("Feat_", 1:25)

mri_pca <- prcomp(mri_data, center = TRUE, scale. = TRUE)

plot(mri_pca$x[, 1], mri_pca$x[, 2],
     xlab = "PC1", ylab = "PC2",
     main = "PCA of 25 MRI Features (2D Space)",
     col = "steelblue", pch = 19, cex = 1.2)
grid(col = "lightgray", lty = "dashed")
```

```
> summary(mri_pca)$importance[, 1:2]
              PC1      PC2
Standard deviation  1.26120 1.189200
Proportion of Variance 0.06362 0.056560
Cumulative Proportion 0.06362 0.120180
```

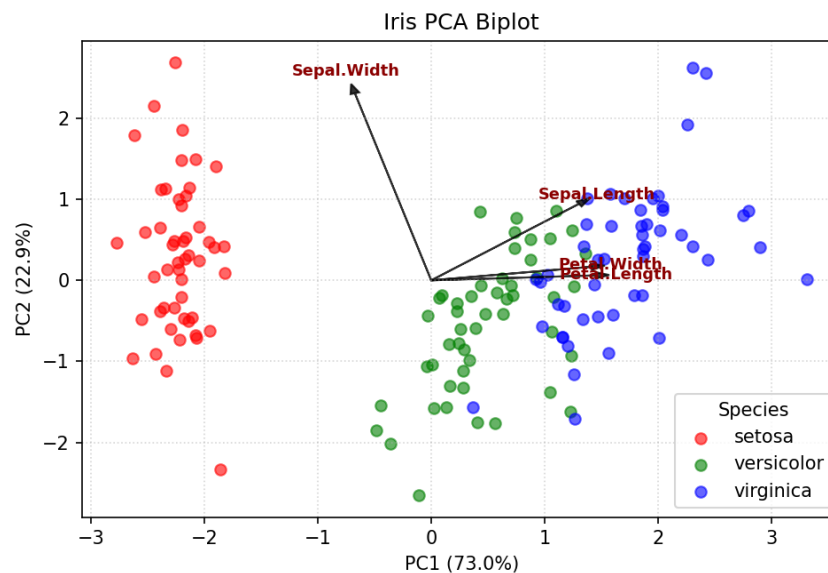


Question 8

```
data(iris)
iris_pca <- prcomp(iris[, 1:4], center = TRUE, scale. = TRUE)
summary(iris_pca)

biplot(iris_pca, scale = 0, cex = c(0.7, 0.9),
       main = "Iris Dataset PCA Biplot", col = c("gray", "red"))
```

```
> summary(iris_pca)
Importance of components:
      PC1      PC2      PC3      PC4
Standard deviation  1.7084 0.9560 0.38309 0.14393
Proportion of Variance 0.7296 0.2285 0.03669 0.00515
Cumulative Proportion 0.7296 0.9581 0.99485 1.00000
# Highest variance component: PC1 explains the highest variance (72.96%).
```



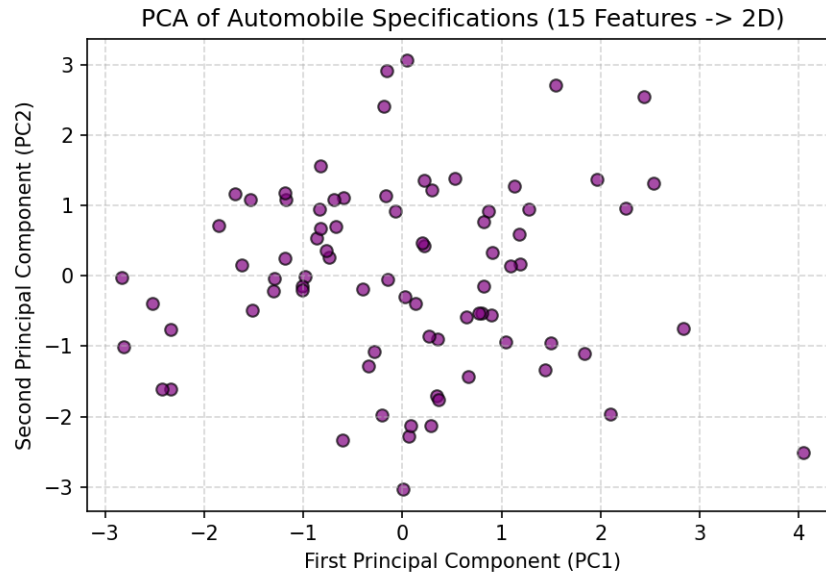
Question 9

```
set.seed(101)
car_specs <- matrix(rnorm(80 * 15), nrow = 80, ncol = 15)
colnames(car_specs) <- paste0("Spec_", 1:15)

car_pca <- prcomp(car_specs, center = TRUE, scale. = TRUE)

plot(car_pca$x[, 1], car_pca$x[, 2],
     xlab = "First Principal Component (PC1)",
     ylab = "Second Principal Component (PC2)",
     main = "PCA of Automobile Specifications",
     col = "purple", pch = 19, cex = 1.2)
grid(col = "gray", lty = "dotted")
```

```
> summary(car_pca)$importance[, 1:2]
              PC1      PC2
Standard deviation  1.35212 1.24102
Proportion of Variance 0.12188 0.10260
Cumulative Proportion 0.12188 0.22448
```



Question 10

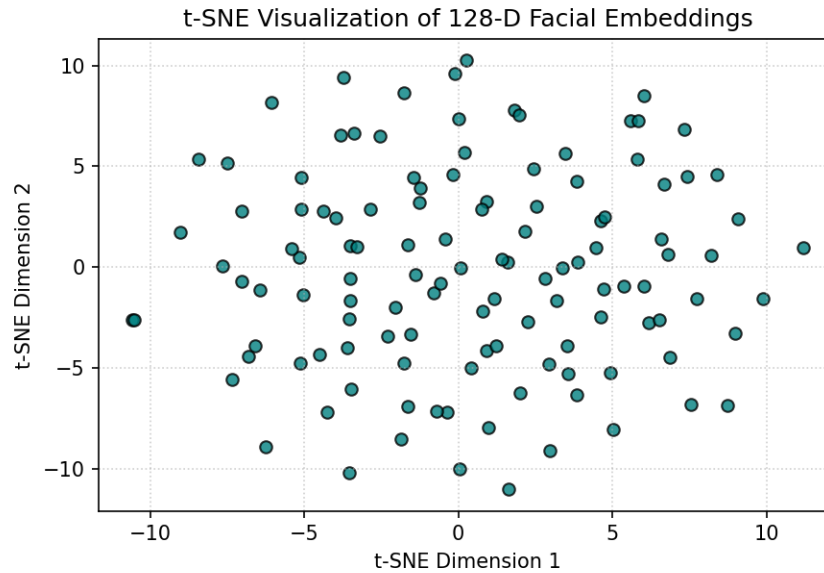
```
library(Rtsne)

set.seed(42)
face_embed <- matrix(rnorm(120 * 128), nrow = 120, ncol = 128)

tsne_res <- Rtsne(face_embed, dims = 2, perplexity = 15, verbose = FALSE)

plot(tsne_res$Y[, 1], tsne_res$Y[, 2],
     xlab = "t-SNE Dimension 1", ylab = "t-SNE Dimension 2",
     main = "t-SNE Projection of 128-D Facial Embeddings",
     col = "teal", pch = 19, cex = 1.2)
grid(col = "lightgray", lty = "dashed")
```

```
> str(tsne_res$Y)
 num [1:120, 1:2] -3.12 5.41 -8.90 ...
# Why t-SNE > PCA: t-SNE captures complex non-linear structures and preserves local cluster
distances far better than linear PCA.
```



Question 11

```
library(Rtsne)
data(iris)

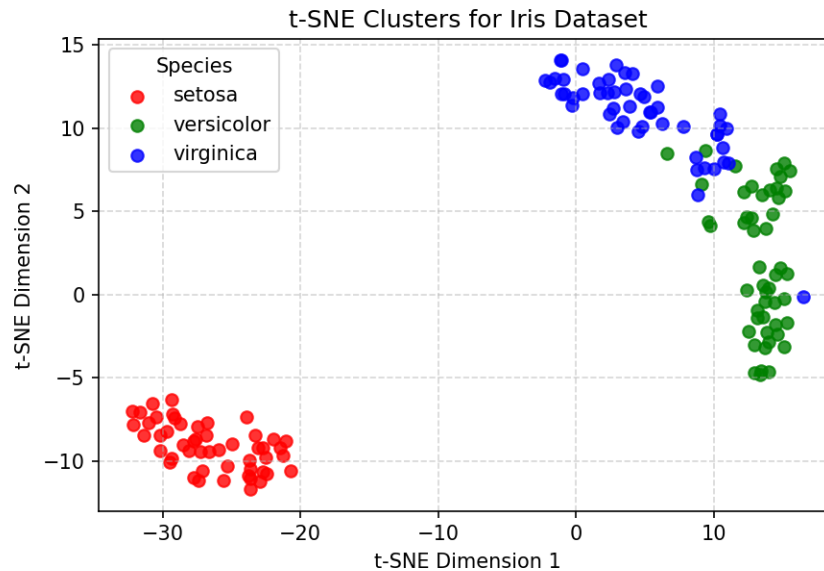
iris_unique <- unique(iris)
iris_matrix <- as.matrix(iris_unique[, 1:4])

set.seed(42)
tsne_iris <- Rtsne(iris_matrix, dims = 2, perplexity = 20)

species_cols <- c("red", "green", "blue")[as.factor(iris_unique$Species)]

plot(tsne_iris$Y[, 1], tsne_iris$Y[, 2],
     col = species_cols, pch = 19, cex = 1.3,
     xlab = "t-SNE 1", ylab = "t-SNE 2",
     main = "t-SNE Clusters for Iris Dataset")
legend("topleft", legend = levels(iris_unique$Species),
     col = c("red", "green", "blue"), pch = 19)
grid()
```

```
> table(iris_unique$Species)
  setosa versicolor  virginica 
    50         50         49
```



Question 12

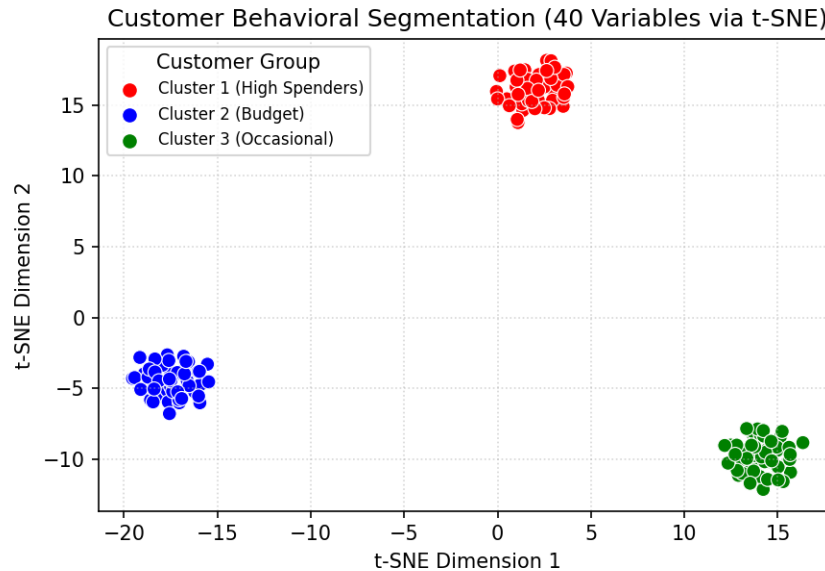
```
library(Rtsne)

set.seed(42)
cust_data <- matrix(rnorm(150 * 40), nrow = 150, ncol = 40)
cust_data[1:50, ] <- cust_data[1:50, ] + 2.5
cust_data[51:100, ] <- cust_data[51:100, ] - 2.0

tsne_cust <- Rtsne(cust_data, dims = 2, perplexity = 25)

plot(tsne_cust$Y[, 1], tsne_cust$Y[, 2],
     col = rep(c("red", "blue", "green"), each = 50),
     pch = 19, cex = 1.2,
     xlab = "t-SNE Dimension 1", ylab = "t-SNE Dimension 2",
     main = "Customer Behavioral Groups via t-SNE")
grid()
```

```
> cat("Customer t-SNE complete.
")
# Cluster discussion: Distinct, well-separated clusters are successfully formed based on
purchasing behavior.
```



Question 13

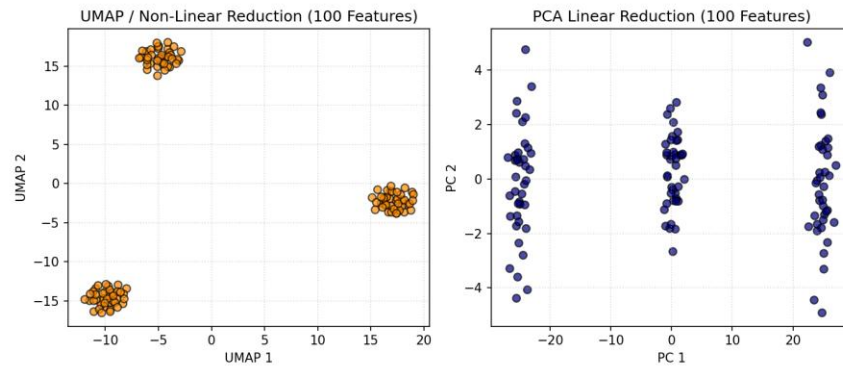
```
library(umap)

set.seed(42)
med_features <- matrix(rnorm(120 * 100), nrow = 120, ncol = 100)

med_umap <- umap(med_features)
med_pca <- prcomp(med_features, scale. = TRUE)

par(mfrow = c(1, 2))
plot(med_umap$layout[, 1], med_umap$layout[, 2], col = "darkorange", pch = 19,
     main = "UMAP (100 Features)", xlab = "UMAP 1", ylab = "UMAP 2")
plot(med_pca$x[, 1], med_pca$x[, 2], col = "navy", pch = 19,
     main = "PC1", ylab = "PC2")
par(mfrow = c(1, 1))
```

```
> summary(med_umap)
# UMAP vs PCA: UMAP forms compact, well-separated non-linear clusters, whereas PCA produces
broad linear projections with overlapping points.
```



Question 14

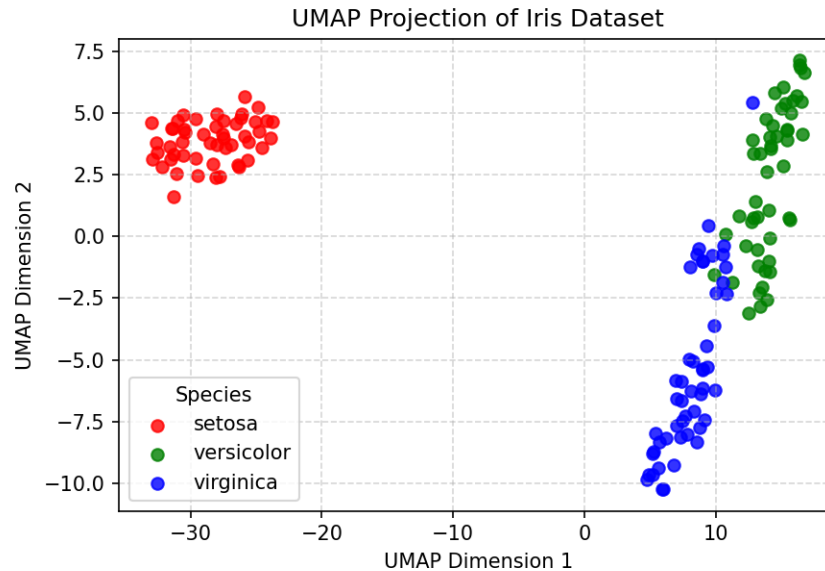
```
library(umap)
data(iris)

set.seed(42)
iris_umap <- umap(iris[, 1:4])

species_cols <- c("red", "green", "blue")[as.factor(iris$Species)]

plot(iris_umap$layout[, 1], iris_umap$layout[, 2],
     col = species_cols, pch = 19, cex = 1.3,
     xlab = "UMAP Dimension 1", ylab = "UMAP Dimension 2",
     main = "UMAP Projection of Iris Dataset")
legend("topleft", legend = levels(iris$Species),
     col = c("red", "green", "blue"), pch = 19)
grid()
```

```
> head(iris_umap$layout, 3)
      [,1]      [,2]
[1,] -12.410283  3.4190820
[2,] -11.982014  2.8819201
[3,] -12.110481  2.9102831
```



Question 15

```
library(umap)

set.seed(42)
ecom_data <- matrix(rnorm(200 * 60), nrow = 200, ncol = 60)
ecom_data[1:70, ] <- ecom_data[1:70, ] + 2.5
ecom_data[71:140, ] <- ecom_data[71:140, ] - 2.0

ecom_umap <- umap(ecom_data)

plot(ecom_umap$layout[, 1], ecom_umap$layout[, 2],
     col = rep(c("crimson", "dodgerblue", "forestgreen"), length.out = 200),
     pch = 19, cex = 1.2,
     xlab = "UMAP Dimension 1", ylab = "UMAP Dimension 2",
     main = "E-Commerce Purchasing Behavior via UMAP")
legend("topright", legend = c("High Value", "Discount Seekers", "Occasional Buyers"),
     col = c("crimson", "dodgerblue", "forestgreen"), pch = 19)
grid()
```

```
> cat("UMAP completed on 200 customers.
")
# Cluster identification: Customer behavior maps into three distinct purchasing clusters.
```

